

BANKING MANAGEMENT SYSTEM

Mini Project Using PostgreSQL

Prepared By:

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Technology Used:

- PostgreSQL
- pgAdmin 4
- SQL

Project Description:

This project demonstrates the implementation of a Banking Management System using PostgreSQL. The database consists of Customers, Accounts, Transactions, and Loans tables. Various SQL operations such as INNER JOIN, LEFT JOIN, GROUP BY, Aggregate Functions, Subqueries, and Window Functions were performed to analyze and manage banking data.

Total Tasks Completed: 17 SQL Operations

Project Highlights

- Database Creation
- Table Creation
- Data Insertion
- Joins (INNER JOIN, LEFT JOIN)
- Aggregate Functions
- Window Functions
- Banking Data Analysis

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File Object Tools Edit View Window Help

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem postgres@PostgreSQL 18*

Query: BankingManagementSystem/postgres@PostgreSQL 18

```
-- ACCOUNTS TABLE--
CREATE TABLE Accounts (
  AccountID INT PRIMARY KEY,
  CustomerID INT,
  Balance NUMERIC(12,2),
  OpenDate DATE,
  FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

--ACCOUNTS TABLE DATA--
INSERT INTO Accounts VALUES
(1001,1,550000,'2025-01-10'),
(1002,2,120000,'2024-11-20'),
(1003,3,350000,'2025-03-15'),
(1004,4,980000,'2025-02-01'),
(1005,5,750000,'2025-01-25'),
(1006,6,1500000,'2024-12-18'),
(1007,7,420000,'2025-04-10'),
(1008,8,880000,'2025-05-05'),
(1009,9,2000000,'2024-09-30'),
(1010,10,670000,'2025-03-22');
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 29, Col 1

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File Object Tools Edit View Window Help

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem postgres@PostgreSQL 18*

Query: BankingManagementSystem/postgres@PostgreSQL 18

```
--LOANS TABLE--
CREATE TABLE Loans (
  LoanID INT PRIMARY KEY,
  CustomerID INT,
  LoanAmount NUMERIC(12,2),
  LoanType VARCHAR(50),
  FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

--LOANS TABLE DATA--
INSERT INTO Loans VALUES
(1,1,500000,'Home Loan'),
(2,2,200000,'Car Loan'),
(3,4,100000,'Education Loan'),
(4,5,300000,'Business Loan'),
(5,6,150000,'Personal Loan'),
(6,8,250000,'Home Loan'),
(7,9,400000,'Business Loan'),
(8,10,180000,'Car Loan');
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 88, Col 21

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Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem

Query: BankingManagementSystem/postgres@PostgreSQL 18

```

1108
1109
1110
1111 SELECT * FROM Customers;
1112 SELECT * FROM Accounts;
1113 SELECT * FROM Transactions;
1114 SELECT * FROM Loans;
1115
-----
1116
1117 --Display customer names, account numbers, and account balances using INNER JOIN--
1118
1119 SELECT c.CustomerName,
1120        c.AccountNo,
1121        a.Balance
1122 FROM Customers c
1123 INNER JOIN Accounts a
1124 ON c.CustomerID = a.CustomerID;
1125
1126 --Top 3 Highest Balance Customers--
1127
1128 SELECT c.CustomerName,
1129        a.Balance
1130 FROM Customers c
1131 FROM Accounts a
1132
Total rows: 10 Query complete 00:00:00.306 CRLF Ln 113, Col 28

```

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Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem

Query: BankingManagementSystem/postgres@PostgreSQL 18

```

1118 --Display customer names, account numbers, and account balances using INNER JOIN--
1119
1120 SELECT c.CustomerName,
1121        c.AccountNo,
1122        a.Balance
1123 FROM Customers c
1124 INNER JOIN Accounts a
1125 ON c.CustomerID = a.CustomerID;
1126
1127 --Top 3 Highest Balance Customers--
1128
1129 SELECT c.CustomerName,
1130        a.Balance
1131 FROM Customers c
1132 JOIN Accounts a
1133 ON c.CustomerID = a.CustomerID
1134 ORDER BY a.Balance DESC
1135 LIMIT 3;
1136
-----
1137
1138 ----Customers with Loan Amount and Loan Type----
1139
1140
Total rows: 10 Query complete 00:00:00.306 CRLF Ln 113, Col 28

```

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Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem

Query: BankingManagementSystem/postgres@PostgreSQL 18

CustomerID	CustomerName	AccountNo	Balance	LoanAmount	LoanType
1	John Doe	1000000001	1000.00	0.00	None
2	Jane Smith	1000000002	2000.00	0.00	None
3	Bob Johnson	1000000003	3000.00	0.00	None
4	Alice Brown	1000000004	4000.00	0.00	None
5	Charlie Davis	1000000005	5000.00	0.00	None
6	Diana Prince	1000000006	6000.00	0.00	None
7	Frank Miller	1000000007	7000.00	0.00	None
8	Grace Wilson	1000000008	8000.00	0.00	None
9	Henry Taylor	1000000009	9000.00	0.00	None
10	Ivy Green	1000000010	10000.00	0.00	None

Total rows: 10 Query complete 00:00:00.170 CRLF Ln 113, Col 1

pgAdmin 4

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem

Query: BankingManagementSystem/postgres@PostgreSQL 18

CustomerID	CustomerName	AccountNo	Balance	LoanAmount	LoanType
1	John Doe	1000000001	1000.00	0.00	None
2	Jane Smith	1000000002	2000.00	0.00	None
3	Bob Johnson	1000000003	3000.00	0.00	None
4	Alice Brown	1000000004	4000.00	0.00	None
5	Charlie Davis	1000000005	5000.00	0.00	None
6	Diana Prince	1000000006	6000.00	0.00	None
7	Frank Miller	1000000007	7000.00	0.00	None
8	Grace Wilson	1000000008	8000.00	0.00	None
9	Henry Taylor	1000000009	9000.00	0.00	None
10	Ivy Green	1000000010	10000.00	0.00	None

Total rows: 10 Query complete 00:00:00.178 CRLF Ln 113, Col 1

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File Object Tools Edit View Window Help

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem postgres Casts Catalogs Event Triggers Extensions Foreign Data Wrappers Languages Publications Schemas Subscriptions Login/Group Roles Tablespaces

Query: BankingManagementSystem/postgres@PostgreSQL 18

```
137 -----
138 -----Customers with Loan Amount and Loan Type-----
139
140
141 SELECT c.CustomerName,
142        l.LoanAmount,
143        l.LoanType
144 FROM Customers c
145 JOIN Loans l
146 ON c.CustomerID = l.CustomerID;
147
148 -----
149
150 --Total Deposit and Withdrawal Amount--
151
152 SELECT TransactionType,
153        SUM(Amount) AS TotalAmount
154 FROM Transactions
155 GROUP BY TransactionType;
156
157 -----
158
159 --Customer-wise Total Transaction Amount--
160
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 113, Col 28

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File Object Tools Edit View Window Help

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) BankingManagementSystem postgres Casts Catalogs Event Triggers Extensions Foreign Data Wrappers Languages Publications Schemas Subscriptions Login/Group Roles Tablespaces

Query: BankingManagementSystem/postgres@PostgreSQL 18

```
158 --Customer-wise Total Transaction Amount--
159
160 SELECT c.CustomerName,
161        SUM(t.Amount) AS TotalTransactionAmount
162 FROM Customers c
163 JOIN Accounts a
164 ON c.CustomerID = a.CustomerID
165 JOIN Transactions t
166 ON a.AccountID = t.AccountID
167 GROUP BY c.CustomerName;
168
169 -----
170 --Customers Having Balance Greater Than Average Balance--
171
172 SELECT c.CustomerName,
173        a.Balance
174 FROM Customers c
175 JOIN Accounts a
176 ON c.CustomerID = a.CustomerID
177 WHERE a.Balance >
178 (
179     SELECT AVG(Balance)
180     FROM Accounts
181 );
182
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 168, Col 1

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

BankingManagementSystem/postgres@PostgreSQL 18

Query Query History

```
183  
184 ---Highest Transaction Amount Per Customer---  
185  
186 SELECT c.CustomerName,  
187        MAX(t.Amount) AS HighestTransaction  
188 FROM Customers c  
189 JOIN Accounts a  
190 ON c.CustomerID = a.CustomerID  
191 JOIN Transactions t  
192 ON a.AccountID = t.AccountID  
193 GROUP BY c.CustomerName;  
194  
195 -----  
196  
197 ---Customers Who Have Not Taken Any Loan--  
198  
199 SELECT c.CustomerName  
200 FROM Customers c  
201 LEFT JOIN Loans l  
202 ON c.CustomerID = l.CustomerID  
203 WHERE l.CustomerID IS NULL;  
204  
205 -----  
206
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 168, Col 1

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

BankingManagementSystem/postgres@PostgreSQL 18

Query Query History

```
206  
207 ---Total Number of Transactions Per Customer---  
208  
209 SELECT c.CustomerName,  
210        COUNT(t.TransactionID) AS TotalTransactions  
211 FROM Customers c  
212 JOIN Accounts a  
213 ON c.CustomerID = a.CustomerID  
214 JOIN Transactions t  
215 ON a.AccountID = t.AccountID  
216 GROUP BY c.CustomerName;  
217  
218 -----  
219 ---Rank Customers by Balance--  
220  
221 SELECT c.CustomerName,  
222        a.Balance,  
223        RANK() OVER(ORDER BY a.Balance DESC) AS RankNo  
224 FROM Customers c  
225 JOIN Accounts a  
226 ON c.CustomerID = a.CustomerID;  
227  
228 -----  
229
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 168, Col 1

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

Servers (1)
 PostgreSQL 18
 Databases (2)
 BankingManagementSystem
 postgres
 Casts
 Catalogs
 Event Triggers
 Extensions
 Foreign Data Wrappers
 Languages
 Publications
 Schemas
 Subscriptions
 Login/Group Roles
 Tablespaces

Query Query History

```

228 -----
229 --Dense Rank Customers by Balance--
230
231
232 SELECT c.CustomerName,
233        a.Balance,
234        DENSE_RANK() OVER(ORDER BY a.Balance DESC) AS DenseRankNo
235 FROM Customers c
236 JOIN Accounts a
237 ON c.CustomerID = a.CustomerID;
238
239 -----
240
241 ---Previous Transaction Amount Using LAG()--
242
243 SELECT TransactionID,
244        Amount,
245        LAG(Amount) OVER(ORDER BY TransactionID) AS PreviousAmount
246 FROM Transactions;
247
248 -----
249
250 ---Next Transaction Amount Using LEAD()--
251
  
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 168, Col 1

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

Servers (1)
 PostgreSQL 18
 Databases (2)
 BankingManagementSystem
 postgres
 Casts
 Catalogs
 Event Triggers
 Extensions
 Foreign Data Wrappers
 Languages
 Publications
 Schemas
 Subscriptions
 Login/Group Roles
 Tablespaces

Query Query History

```

279 --Customers Who Performed More Than 2 Transactions--
280
281 SELECT c.CustomerName,
282        COUNT(t.TransactionID) AS TransactionCount
283 FROM Customers c
284 JOIN Accounts a
285 ON c.CustomerID = a.CustomerID
286 JOIN Transactions t
287 ON a.AccountID = t.AccountID
288 GROUP BY c.CustomerName
289 HAVING COUNT(t.TransactionID) > 2;
290
291 -----
292 --Customer-wise Minimum and Maximum Transaction Amount--
293
294 SELECT c.CustomerName,
295        MIN(t.Amount) AS MinimumTransaction,
296        MAX(t.Amount) AS MaximumTransaction
297 FROM Customers c
298 JOIN Accounts a
299 ON c.CustomerID = a.CustomerID
300 JOIN Transactions t
301 ON a.AccountID = t.AccountID
302 GROUP BY c.CustomerName;
  
```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 289, Col 35

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

BankingManagementSystem/postgres@PostgreSQL 18

BankingManagementSystem/postgres

customername character varying (100) minimumtransaction numeric maximumtransaction numeric

1	Rohit Kumar	9000.00	15000.00
2	Karan Mehta	3500.00	12000.00
3	Neha Joshi	7000.00	7000.00
4	Sneha Patil	10000.00	25000.00
5	Aman Verma	3000.00	5000.00
6	Rahul Sharma	5000.00	10000.00
7	Vivek Shah	30000.00	30000.00
8	Pooja Sharma	4500.00	4500.00
9	Anjali Verma	2000.00	2000.00
10	Priya Singh	15000.00	15000.00

Total rows: 10 Query complete 00:00:00.123 CRLF Ln 291, Col 1

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

BankingManagementSystem/postgres@PostgreSQL 18

BankingManagementSystem/postgres

customername character varying (100) totaltransactions bigint

1	Rohit Kumar	2
2	Karan Mehta	2
3	Neha Joshi	1
4	Sneha Patil	2
5	Aman Verma	2
6	Rahul Sharma	2
7	Vivek Shah	1
8	Pooja Sharma	1
9	Anjali Verma	1
10	Priya Singh	1

Total rows: 10 Query complete 00:00:00.216 CRLF Ln 207, Col 1

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

Servers (1)
PostgreSQL 18
Databases (2)
BankingManagementSystem

Query Query History

```

255 FROM Transactions;
256
257 -----
258
259 --Running Total Using SUM() OVER()--
260
261 SELECT TransactionID,
262        Amount,
263        SUM(Amount) OVER(ORDER BY TransactionID) AS RunningTotal
264 FROM Transactions;
265
266 -----
267
268 --Second Highest Account Balance--
269
270 SELECT MAX(Balance) AS SecondHighestBalance
271 FROM Accounts
272 WHERE Balance <
273 (
274     SELECT MAX(Balance)
275     FROM Accounts
276 );
277

```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 289, Col 35

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File Object Tools Edit View Window Help

Object E: BankingManagementSystem/postgres@PostgreSQL 18*

Servers (1)
PostgreSQL 18
Databases (2)
BankingManagementSystem

Query Query History

```

240
241 ---Previous Transaction Amount Using LAG()--
242
243 SELECT TransactionID,
244        Amount,
245        LAG(Amount) OVER(ORDER BY TransactionID) AS PreviousAmount
246 FROM Transactions;
247
248 -----
249
250 ---Next Transaction Amount Using LEAD()--
251
252 SELECT TransactionID,
253        Amount,
254        LEAD(Amount) OVER(ORDER BY TransactionID) AS NextAmount
255 FROM Transactions;
256
257 -----
258
259 --Running Total Using SUM() OVER()--
260
261 SELECT TransactionID,
262        Amount,
263        SUM(Amount) OVER(ORDER BY TransactionID) AS RunningTotal

```

Total rows: 10 Query complete 00:00:00.306 CRLF Ln 289, Col 35